SQL Joins (Inner, Left, Right and Full Join)

SQL Joins are used to combine rows from two or more tables based on a related column between them. They help in retrieving connected data stored across multiple tables.

Types of SQL Joins

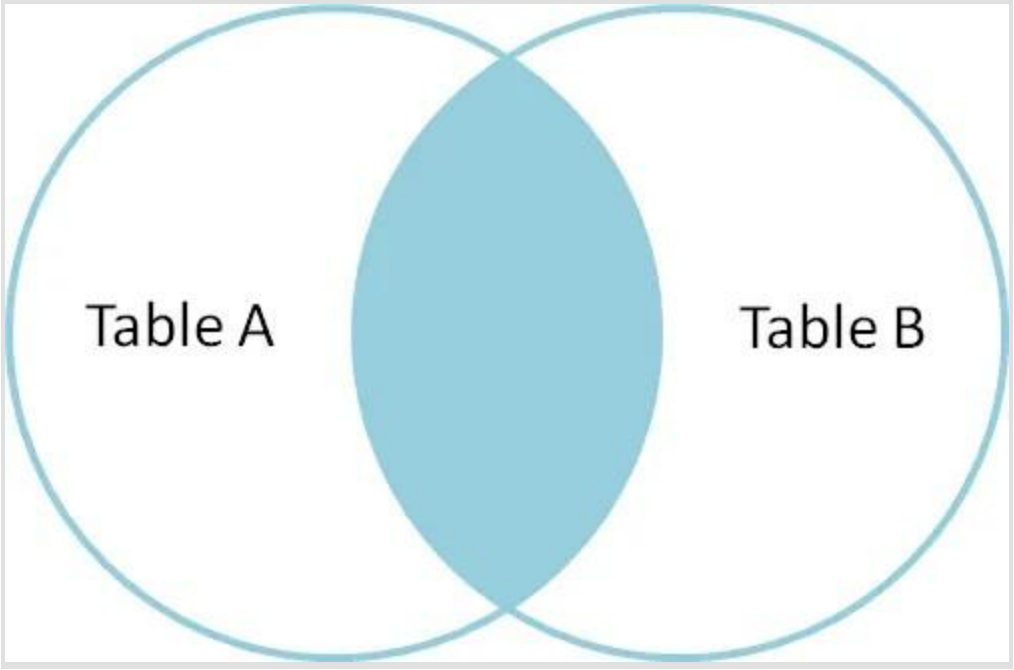
SQL joins are categorized into different types based on how rows from two tables are matched and combined.

1. SQL INNER JOIN

The INNER JOIN in SQL returns only the rows where there is a matching value in both tables. It combines records based on a related column and excludes non-matching rows.

Syntax:

```
SELECT table1.column1,table1.column2,table2.column1,.... FROM table1  INNER JOIN

table2 ON  table1.matching_column = table2.matching_column;
```



Inner Join

***Note**: We can also write JOIN instead of INNER JOIN. JOIN is same as INNER JOIN.*

Example of INNER JOIN:

Consider the two tables, Student and StudentCourse, which share a common column ROLL_NO. Using SQL JOINS, we can combine data from these tables based on their relationship, allowing us to retrieve meaningful information like student details along with their enrolled courses.

S_ID	NAME	ADDRESS
1	Liam	London
2	Sofia	Berlin
3	Mateo	Madrid
4	Emma	Paris
5	Noah	Rome

Students Table

COURSE_ID	ROLL_NO
1	1
2	2
3	3
6	7

StudentCourse Table

Let's look at the example of INNER JOIN clause, and understand it's working. This query will show the names and age of students enrolled in different courses.

Query:

```
SELECT StudentCourse.COURSE_ID, Student.NAME, Student.AGE FROM Student

INNER JOIN StudentCourse
```

```
ON Student.ROLL_NO = StudentCourse.ROLL_NO;
```

Output:

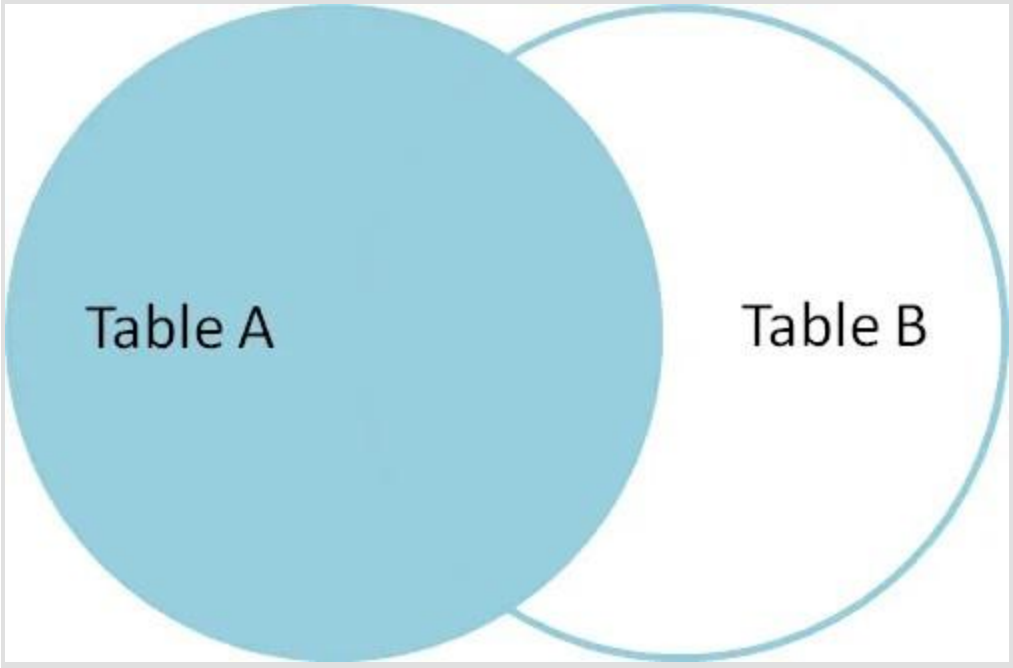
COURSE_ID	NAME	AGE
1	Liam Johnson	18
2	Emma Wilson	19
3	Noah Smith	20

2. SQL LEFT JOIN

A LEFT JOIN returns all rows from the left table, along with matching rows from the right table. If there is no match, NULL values are returned for columns from the right table. LEFT JOIN is also known as LEFT OUTER JOIN.

Syntax:

```
SELECT table1.column1,table1.column2,table2.column1,....

FROM table1

LEFT JOIN table2

ON table1.matching_column = table2.matching_column;
```



Left JOIN

***Note**: We can also use LEFT OUTER JOIN instead of LEFT JOIN, both are the same.*

Example: In this example, the LEFT JOIN retrieves all rows from the Student table and the matching rows from the StudentCourse table based on the ROLL_NO column.

Query:

```
SELECT Student.NAME,StudentCourse.COURSE_ID

FROM Student
```

```
LEFT JOIN StudentCourse

ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

Output:

NAME	COURSE_ID
Liam Johnson	1
Emma Wilson	2
Noah Smith	3
Olivia Brown	
Ava Davis	

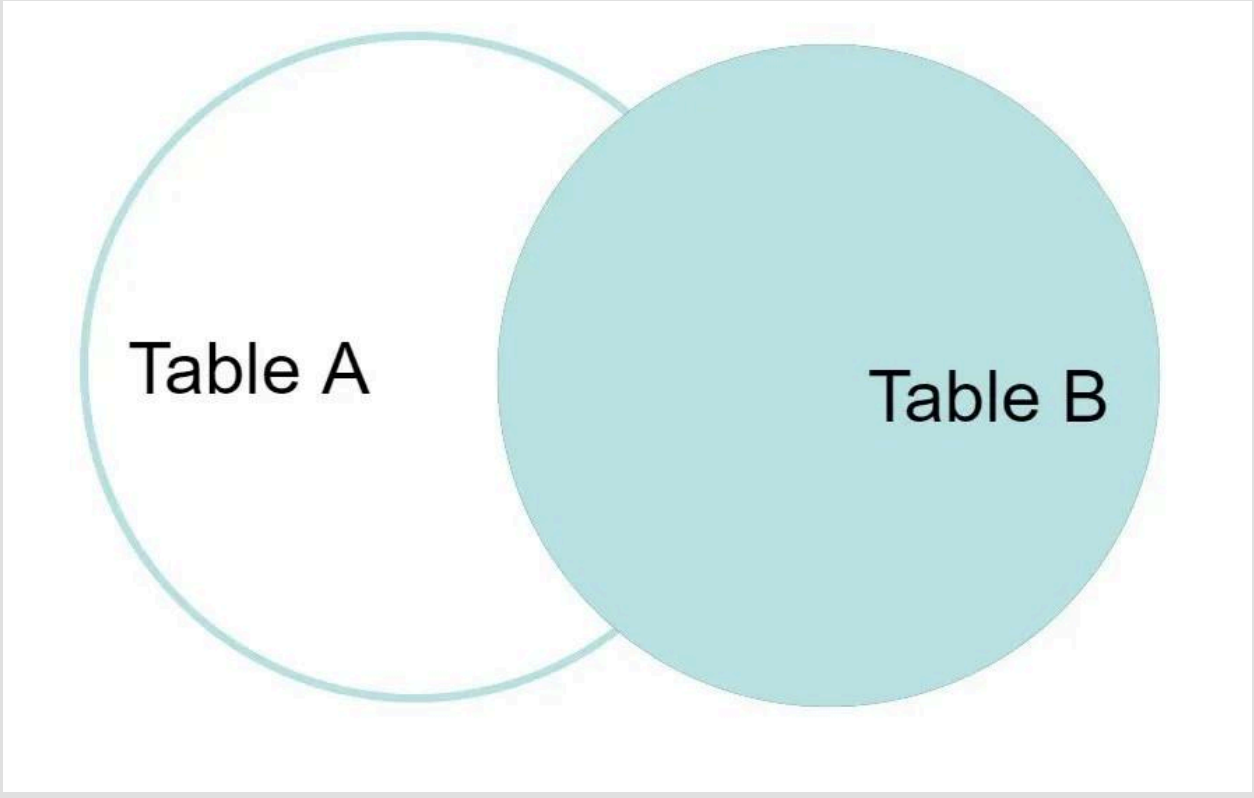
3. SQL RIGHT JOIN

RIGHT JOIN returns all the rows of the table on the right side of the join and matching rows for the table on the left side of the join. It is very similar to LEFT JOIN for the rows for which there is no matching row on the left side, the result-set will contain null. RIGHT JOIN is also known as RIGHT OUTER JOIN.

Syntax

```
SELECT table1.column1,table1.column2,table2.column1,....

FROM table1

RIGHT JOIN table2
```

ON table1.matching_column = table2.matching_column;



Right JOIN

***Note**: We can also use RIGHT OUTER JOIN instead of RIGHT JOIN, both are the same*

Example: In this example, the RIGHT JOIN retrieves all rows from the StudentCourse table and the matching rows from the Student table based on the `ROLL_NO` column.

Query:

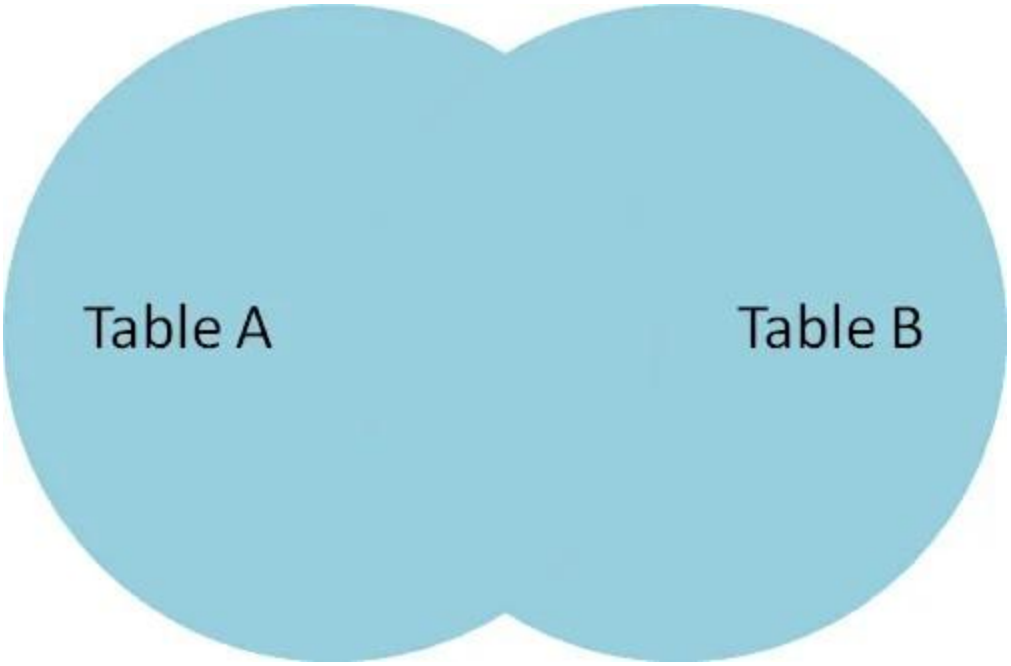
```
SELECT Student.NAME,StudentCourse.COURSE_ID

FROM Student

RIGHT JOIN StudentCourse

ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

Output:

NAME	COURSE_ID
Liam Johnson	1
Emma Wilson	2
Noah Smith	3
	6

4. SQL FULL JOIN

FULL JOIN creates the result-set by combining results of both LEFT JOIN and RIGHT JOIN. The result-set will contain all the rows from both tables. For the rows for which there is no matching, the result-set will contain NULL values.



Syntax

```
SELECT table1.column1,table1.column2,table2.column1,....

FROM table1

FULL JOIN table2

ON table1.matching_column = table2.matching_column;
```

Example: This example uses a FULL JOIN (simulated in SQLite) to return all rows from both tables. Matching records appear together, while non-matching records still show up with NULL values for the missing fields.

Query:

```
SELECT Student.NAME,StudentCourse.COURSE_ID

FROM Student

FULL JOIN StudentCourse

ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

Output :

NAME	COURSE_ID
Liam Johnson	1
Emma Wilson	2
Noah Smith	3
Olivia Brown	
Ava Davis	
	6

5. SQL Natural Join

A Natural Join is a type of INNER JOIN that automatically joins two tables based on columns with the same name and data type. It returns only the rows where the values in the common columns match.

- It joins tables using common columns with the same name.
- It returns only rows where values in those columns match.
- The common column appears only once in the result.

Example: Look at the two tables below. Employee **Table:**

Emp_id	Emp_name	Dept_id
1	Ram	10
2	Jon	30
3	Bob	50